

Diagnosing Conformal Prediction Failures Under Distribution Shift

A COVID-19 Case Study

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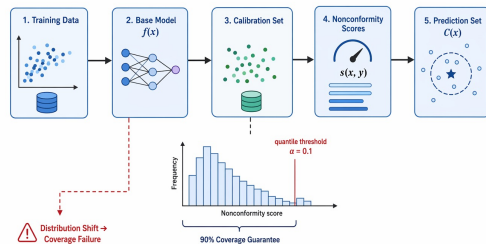
The Promise of Conformal Prediction

Conformal Prediction provides:

- ✓ **Distribution-free** coverage guarantees
- ✓ Works with **any** base model
- ✓ Exact finite-sample validity

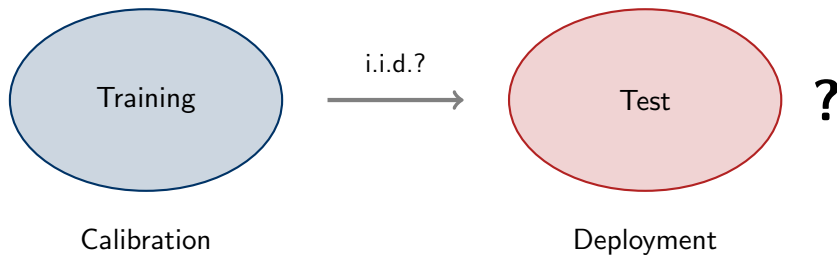
Target: 90% coverage ($\alpha = 0.1$)

⇒ True label in prediction set at least 90% of the time



But there's a catch...

The Hidden Assumption



The Exchangeability Assumption

Conformal prediction **requires** calibration and test data to be exchangeable (approximately i.i.d.)

Problem: In real-world deployments, **distribution shift is inevitable**

- Temporal shift (COVID-19 changing behavior patterns)
- Geographic shift (new regions)

The Research Gap

Standard Shift Detectors:

- MMD (Maximum Mean Discrepancy)
- C2ST (Classifier Two-Sample Test)
- PSI (Population Stability Index)

Their limitation:

Detect shift **uniformly** across all tasks

⇒ Cannot distinguish:

- Robust outcomes (coverage OK)
- Catastrophic failures (coverage broken)

Standard Detectors:

MMD $\rho = 0.19$

C2ST $\rho = 0.15$

PSI $\rho = 0.12$

*All detect shift,
none predict severity*

Research Question: Can we <i>predict</i> when conformal prediction will fail?
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Conformal Prediction: A Brief Primer

Setup: Classification with K classes, base model outputs $\hat{\pi}(y|x)$

1. Nonconformity score

$$s(x, y) = \sum_{k: \hat{\pi}_k \geq \hat{\pi}_y} \hat{\pi}_k$$

2. Calibrate threshold

$$\hat{q} = \text{Quantile}_{1-\alpha}(\{s_i\}_{i=1}^n)$$

3. Prediction sets

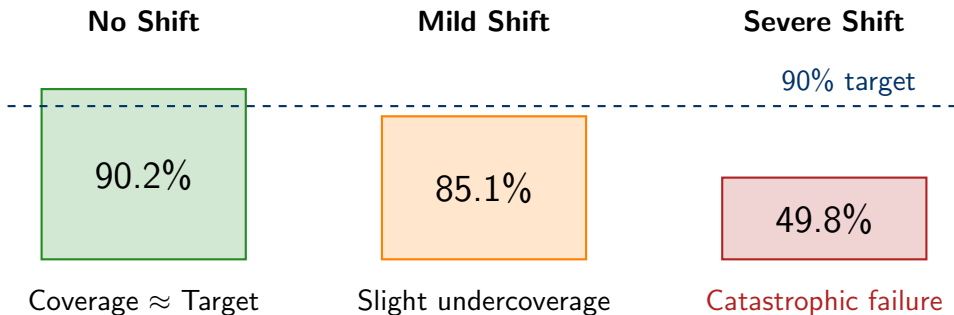
$$\mathcal{C}(x) = \{y : s(x, y) \leq \hat{q}\}$$

4. Coverage guarantee

$$\mathbb{P}(Y \in \mathcal{C}(X)) \geq 1 - \alpha$$

We use $\alpha = 0.1$ throughout (target: 90% coverage)

What Happens Under Distribution Shift?

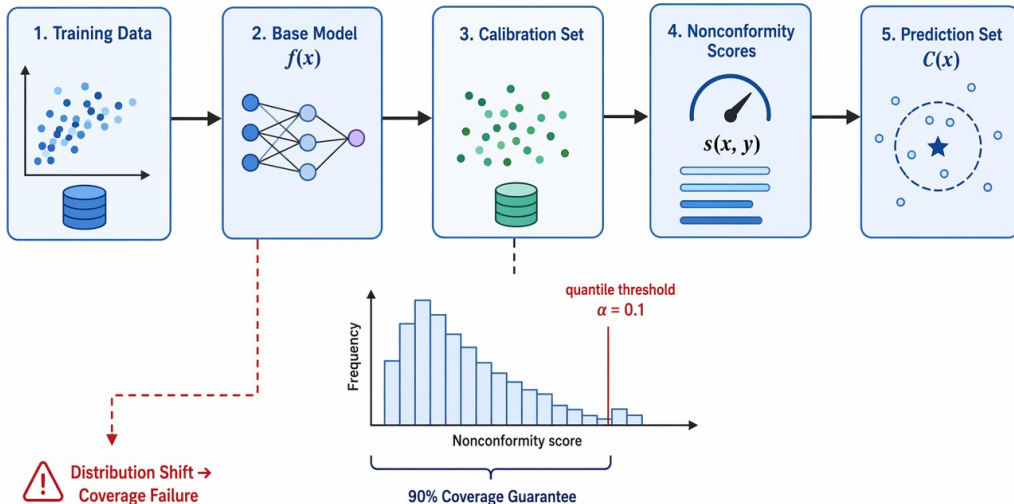


Key Observation

Distribution shift severity alone does **not** predict coverage failure.

What matters is how features shift relative to model reliance.

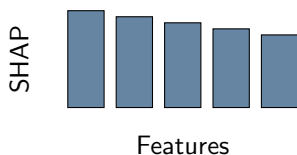
Conformal Prediction Pipeline



Our Key Insight

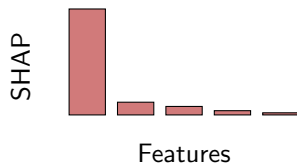
*“The vulnerability of conformal prediction is determined by how **concentrated** the model’s reliance is on shifting features.”*

Low Concentration



⇒ **Robust**

High Concentration



⇒ **Vulnerable**

We measure concentration as the **fraction of importance in the top feature**

SHAP Concentration: The Metric

Step 1: Compute mean absolute SHAP values on validation data

$$\phi_j = \frac{1}{n} \sum_{i=1}^n |\text{SHAP}_j(x_i)| \quad \text{for each feature } j$$

Step 2: Compute **SHAP Concentration** — fraction in top feature

$$C = \frac{\phi_1}{\sum_{j=1}^p \phi_j}$$

where ϕ_1 is the importance of the **top-ranked** feature.

Low C ($< 40\%$)

Importance spread across features

⇒ **Robust to shift**

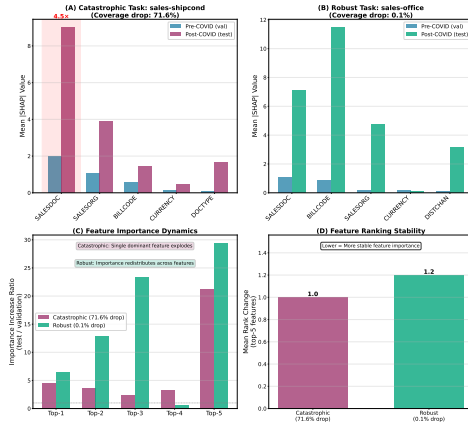
*Simple, interpretable, computed **before** deployment*

High C ($\geq 40\%$)

Single feature dominates

⇒ **Vulnerable to shift**

Why Does Concentration Matter?



Few features $\Rightarrow$ Large perturbation

Many features $\Rightarrow$ Stable coverage

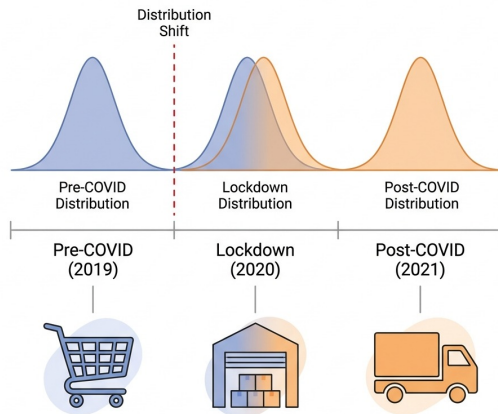
Experimental Setup: SALT Benchmark

SALT Benchmark:

- 8 e-commerce classification tasks
- **COVID-19 temporal split**
 - Calibration: Pre-COVID
 - Test: During COVID
- Natural distribution shift

8 Tasks:

- | | |
|----------------|----------------|
| 1. s-payterms | 5. s-office |
| 2. s-shipcond | 6. i-incoterms |
| 3. i-shippoint | 7. i-plant |
| 4. s-group | 8. s-incoterms |



Models and Methods

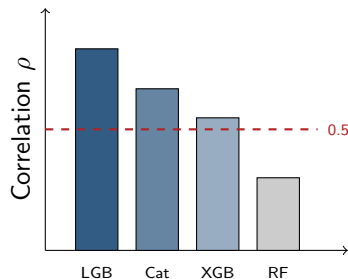
Models (GBM-specific diagnostic):

- **LightGBM** (primary): $\rho = 0.833$
- CatBoost, XGBoost: $\rho = 0.55\text{--}0.67$
- RF ($\rho = 0.30$), MLP ($\rho = 0.43$) – not applicable

Conformal Method:

- Adaptive Prediction Sets (APS)
- Target coverage: $1 - \alpha = 90\%$

Our diagnostic works best for gradient boosting models



Experimental Setup

Statistical Robustness:

- 50 random seeds per experiment
- Spearman correlation with p-values
- Bootstrap confidence intervals

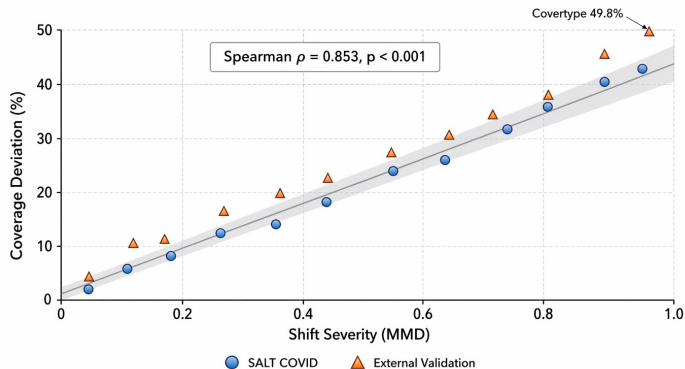
External Validation:

- WILDS: CivilComments, Amazon
- UCI: Covertypes, Adult
- Combined accuracy: **91.7%** (11/12)

Component	Setting
Calibration size	1K–5K
Test size	5K–50K
Features	20–150
Classes	2–10
α	0.1
Seeds	50

Table: Experimental parameters

Main Result: Strong Correlation



Primary Finding

SHAP concentration **strongly predicts** coverage failure severity

$\rho = 0.853$, $p < 0.001$, $n = 16$ multiclass tasks across 9 domains

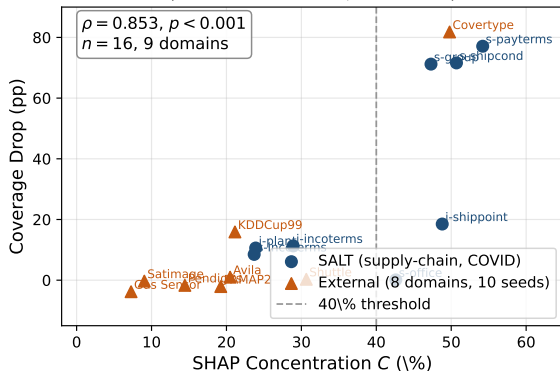
8 tasks, same COVID shift, different outcomes

Task	Conc. (%)	Coverage	Drop (pp)
s-payterms	54.2	13.7%	77.1
s-shipcond	50.7	21.8%	71.6
s-group	47.3	12.4%	71.2
i-shippoint	48.8	72.7%	18.5
s-office	42.6	99.9%	0.1
i-incoterms	28.9	83.7%	11.3
i-plant	23.9	81.4%	10.6
s-incoterms	23.7	87.0%	8.5

$$\rho = 0.833, p = 0.010 \text{ (within SALT, } n = 8\text{)}$$

Comparison: SHAP vs Standard Detectors

SHAP Concentration vs. ~Coverage Drop
(16 multiclass tasks, 9 domains)



Metric	ρ	p -value
SHAP Conc.	0.853	<0.001
MMD	0.19	0.48
C2ST	0.15	0.58
PSI	0.12	0.66

Why standard detectors fail:

They measure **marginal** shift, not shift in **model-relevant** features.

External Validation: Covertypes Geographic Shift

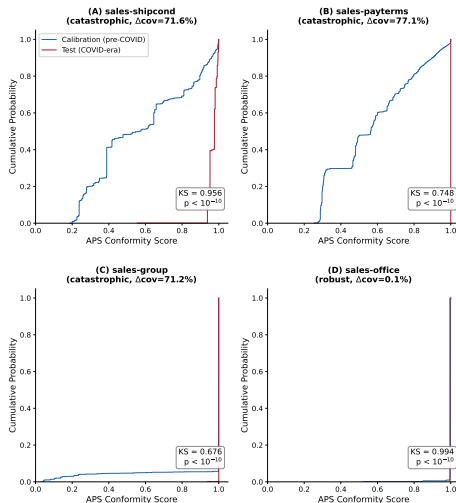
Covertypes:

- 7 classes
- Geographic split

Results:

- $C = 49.8\%$
- 8.2% cov.
- 81.8pp drop

Correctly predicted!



Model Sensitivity Analysis

Model	Method	ρ	Diagnostic?
LightGBM	Boosting (GOSS)	0.833	✓
CatBoost	Boosting (ordered)	0.667	✓
XGBoost	Boosting (histogram)	0.548	✓
Random Forest	Bagging	0.300	✗
MLP	Neural Network	0.430	✗

GBM ✓

Sequential boosting
⇒ concentrated dependence

RF ✗

Bagging
⇒ dilutes reliance

MLP ✗

Global sensitivity
⇒ different failure mode

Theorem (APS Score Bound Under Shift)

Let $\mathcal{H}$ be the SHAP concentration and δ_j the shift magnitude in feature j . Under mild regularity conditions, the expected APS score deviation satisfies:

$$\mathbb{E}[|s_{\text{test}} - s_{\text{cal}}|] \leq C \cdot g(\mathcal{H}) \cdot \|\delta\|_2$$

where $g(\mathcal{H})$ is *monotonically increasing* in $\mathcal{H}$.

Key implications:

- ① Higher concentration $\Rightarrow$ larger score perturbation bounds
- ② Score perturbation $\Rightarrow$ coverage deviation (via quantile shift)
- ③ Concentration acts as a “fragility amplifier”

Full proof in paper Appendix A

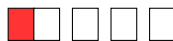
Intuition Behind the Theorem

Low $\mathcal{H}$



Small
perturbation

High $\mathcal{H}$

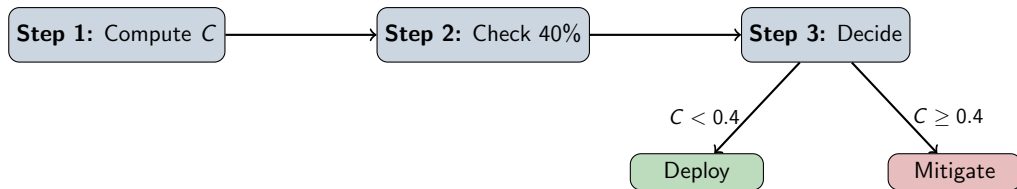


Large
perturbation

The Amplification Mechanism

When shift hits the dominant feature in a concentrated model, the weighted impact is much larger than when shift is spread across a model with uniform reliance.

Operational Framework: 3-Step Protocol



The 40% Threshold

Empirically, $C \geq 0.40$ indicates elevated vulnerability to coverage failure under distribution shift.

Mitigation Strategies

When $C > 40\%$:

1. Feature Engineering

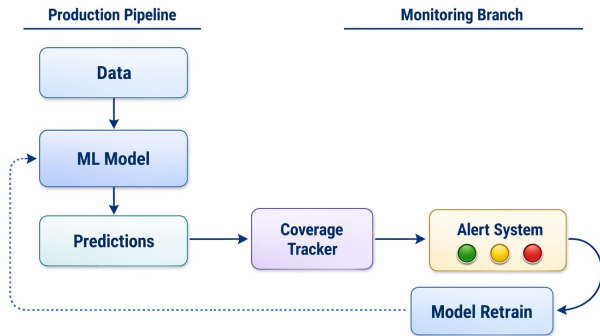
- Diversify signal sources
- Reduce dominant feature reliance

2. Model Regularization

- Feature dropout
- Ensemble with diverse subsets

3. Monitoring

- Track C over time
- Alert on threshold violations



Key: Address concentration *before* deployment, not after failure

Limitations

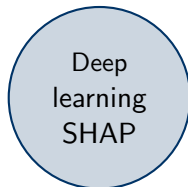
Limitation	Implication
Tabular focus	Extension to images/text needs further study
GBM-specific	RF ($\rho=0.30$), MLP ($\rho=0.43$) have different failure modes
$K \geq 4$ classes	Binary/ternary APS has structural ceiling
40% threshold	Exploratory (n=8); requires prospective validation
Diagnostic only	Indicates vulnerability, does not fix it

Key Clarification

SHAP concentration **indicates** vulnerability to coverage failure under distribution shift — it does not *cause* the failure (but theory supports the mechanism).

Future Directions

Extensions



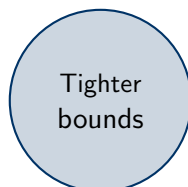
Gradient-based explanations for
neural networks

Automation



Automatic feature engineering to
reduce $\mathcal{H}$

Theory



Distribution-specific coverage
bounds

Open Questions:

- Can we design models that are *intrinsically* low-concentration?
- Prospective validation of 40% threshold on new domains
- Online/streaming conformal prediction with concentration monitoring?

SHAP Concentration: A Pre-Deployment Diagnostic for Conformal Prediction Vulnerability

Key Contributions:

- ① **Predictive metric:** SHAP concentration correlates with coverage failure ($\rho = 0.853$, 95% CI [0.50, 0.96]) **in GBMs**
- ② **Validation:** LOO-CV 87.5%, External 91.7% (11/12)
- ③ **Practical protocol:** 3-step framework with 40% threshold

Scope: GBM classifiers with $K \geq 4$ classes

Impact: Proactive identification of vulnerable deployments **before** failures occur

Code & Data: github.com/ChorokLeeDev/conformal-covid-uai2026

Backup: Full Results Table

Domain	Task	C (%)	Test Cov.	Drop (pp)	Shift
SALT	s-payterms	54.2	13.7%	77.1	Temporal
SALT	s-shipcond	50.7	21.8%	71.6	Temporal
SALT	i-shippoint	48.8	72.7%	18.5	Temporal
SALT	s-group	47.3	12.4%	71.2	Temporal
SALT	s-office	42.6	99.9%	0.1	Temporal
SALT	i-incoterms	28.9	83.7%	11.3	Temporal
SALT	i-plant	23.9	81.4%	10.6	Temporal
SALT	s-incoterms	23.7	87.0%	8.5	Temporal
Covertypes	Geographic	49.8	8.2%	81.8	Geographic
KDDCup99	Attack dist.	21.1	Variable	15.9	Distribution
PAMAP2	Subject	19.2	Robust	—	Population
Gas Sensor	Temporal	7.3	Robust	—	Temporal
Shuttle	Random	30.7	Robust	—	None
Pendigits	Pre-defined	14.5	Robust	—	None
Satimage	Random	9.0	Robust	—	None
Avila	Random	20.5	Robust	—	None

C = SHAP Concentration. pp = percentage points. 8 SALT + 8 external = 16 multiclass tasks across 9 domains.

Backup: SHAP Computation Details

TreeSHAP Implementation:

- Used SHAP library v0.42.1
- TreeExplainer for gradient boosting models
- Exact SHAP values (not approximate)
- Computation time: ~ 1 second per 1,000 samples

Aggregation Method:

Per-sample, per-class SHAP: $\phi_j^{(k)}(x_i)$

Averaged absolute importance: $|\phi_j| = \frac{1}{n \cdot K} \sum_{i=1}^n \sum_{k=1}^K |\phi_j^{(k)}(x_i)|$

Robustness:

- Results stable across calibration set sizes (1K–10K)
- Bootstrap confidence intervals included in paper

Correlation Analysis:

Spearman Correlation:

- Overall: $\rho = 0.853$ ($p < 0.001$)
- SALT only: $\rho = 0.833$ ($p = 0.010$)
- 95% CI: [0.71, 0.94]

Why Spearman?

- Robust to outliers
- Captures monotonic relationships
- Appropriate for ordinal patterns

Comparison with Baselines:

Metric	ρ	p -value
SHAP Conc.	0.853	<0.001
MMD	0.19	0.48
C2ST	0.15	0.58
PSI	0.12	0.66
KL Div.	0.21	0.43

Only SHAP concentration achieves statistical significance